Enrollment No:-240840116014

Practical 3: Apply fltering, sorting, aggregation, and GroupBy operations on real-world datasets.

Step 1: Import Pandas and Create Dataset

```
# Import the pandas library
import pandas as pd
# Create a sample employee dataset
Data = {
    "Emp_ID":[101,102,103,104,105,106,107,108],
    "Name": ["Rahul","Priya","Amit","Riya","karan","Neha","Vivek","Pooja"],
    "Department":["IT","HR","IT","Finance","HR","Finance","IT","HR"],
    "City":["Surat","Navsari","Surat","Valsad","Surat","Navsari","Valsad","Surat"],
    "Salary":[45000,38000,52000,48000,40000,51000,60000,42000],
    "Experience":[2,1,5,4,2,6,8,3]
}
# Convert the dictionary into a DataFrame
df = pd.DataFrame(Data)
# Display the dataset
print(df)
```

```
   Emp_ID   Name Department     City  Salary  Experience
0     101  Rahul         IT    Surat   45000           2
1     102  Priya         HR  Navsari   38000           1
2     103   Amit         IT    Surat   52000           5
3     104   Riya    Finance   Valsad   48000           4
4     105  karan         HR    Surat   40000           2
5     106   Neha    Finance  Navsari   51000           6
6     107  Vivek         IT   Valsad   60000           8
7     108  Pooja         HR    Surat   42000           3
```

```
#Display the Dataset
df
```

	Emp_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
1	102	Priya	HR	Navsari	38000	1
2	103	Amit	IT	Surat	52000	5
3	104	Riya	Finance	Valsad	48000	4
4	105	karan	HR	Surat	40000	2
5	106	Neha	Finance	Navsari	51000	6
6	107	Vivek	IT	Valsad	60000	8
7	108	Pooja	HR	Surat	42000	3

Step 2: Filter Data

```
#Display Employees from IT Department
df[df["Department"]=="IT"]
```

	Emp_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

```
#Display Employees with salary Greater than 50,000
df[df["Salary"]>50000]
```

	Emp_ID	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navsari	51000	6
6	107	Vivek	IT	Valsad	60000	8

```
#Display Employees from Surat
df[df["City"]=="Surat"]
```

	Emp_ID	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
4	105	karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3

```
#Apply Multiple Conditions
df[(df["Department"]=="IT")& (df["Salary"]>50000)]
```

	Emp_ID	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

Step 3: Sorting

```
#Sort values by salary
df.sort_values(by="Salary")
```

	Emp_ID	Name	Department	City	Salary	Experience
1	102	Priya	HR	Navsari	38000	1
4	105	karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3
0	101	Rahul	IT	Surat	45000	2
3	104	Riya	Finance	Valsad	48000	4
5	106	Neha	Finance	Navsari	51000	6
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

```
#Sort Salary in Descending Order
df.sort_values(by="Salary", ascending=False)
```

	Emp_ID	Name	Department	City	Salary	Experience
6	107	Vivek	IT	Valsad	60000	8
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navsari	51000	6
3	104	Riya	Finance	Valsad	48000	4
0	101	Rahul	IT	Surat	45000	2
7	108	Pooja	HR	Surat	42000	3
4	105	karan	HR	Surat	40000	2
1	102	Priya	HR	Navsari	38000	1

```
#Sort by Department and Salary
df.sort_values(by=["Department","Salary"], ascending=[True,False])
```

	Emp_ID	Name	Department	City	Salary	Experience
5	106	Neha	Finance	Navsari	51000	6
3	104	Riya	Finance	Valsad	48000	4
7	108	Pooja	HR	Surat	42000	3
4	105	karan	HR	Surat	40000	2
1	102	Priya	HR	Navsari	38000	1
6	107	Vivek	IT	Valsad	60000	8
2	103	Amit	IT	Surat	52000	5
0	101	Rahul	IT	Surat	45000	2

Step 4:Perform Aggregation

```
#Calculate Total Salary
df["Salary"].sum()
```

```
np.int64(376000)
```

```
#Calculate Average Salary
df["Salary"].mean()
```

```
np.float64(47000.0)
```

```
#Find Highest Salary
df["Salary"].max()
```

```
60000
```

```
#Find Lowest Salary
df["Salary"].min()
```

```
38000
```

```
#Count Employees
df["Emp_ID"].count()
```

```
np.int64(8)
```

Step 5: Perform Groupby Operation

```
#Average Salary Department-Wise
df.groupby("Department")["Salary"].mean()
```

Department	Salary
Finance	49500.000000
HR	40000.000000
IT	52333.333333

dtype: float64

```
#minimum salary department-wise
df.groupby("Department")["Salary"].min()
```

Department	Salary
Finance	48000
HR	38000
IT	45000

dtype: int64

```
#Count the total Number of Employees Department-wise
df.groupby("Department")["Emp_ID"].count()
```

Department	Emp_ID
Finance	2
HR	3
IT	3

dtype: int64

```
#Total Salary Department-wise
df.groupby("Department")["Salary"].sum()
```

Department	Salary
Finance	99000
HR	120000
IT	157000

dtype: int64

```
#Maximum salary Department-wise
df.groupby("Department")["Salary"].max()
```

Department	Salary
Finance	51000
HR	42000
IT	60000

dtype: int64

```
#Count salary Department wise
df.groupby("Department")["Salary"].count()
```

Department	Salary
Finance	2
HR	3
IT	3

dtype: int64

Step 6: Multiple Aggregation Functions

```
## Calculate multiple salary statistics for each department
df.groupby("Department")["Salary"].agg(["count","sum","mean","min","max"])
```

Department	count	sum	mean	min	max
Finance	2	99000	49500.000000	48000	51000
HR	3	120000	40000.000000	38000	42000
IT	3	157000	52333.333333	45000	60000

Step 7: Save the Result

```
#Save file as csv and Download
df.to_csv("Employee_Data.csv", index=False)
from google.colab import files
files.download("Employee_Data.csv")
```